

8 (short)

(a) Describe the operation of a tri-state buffer and explain how this is used to facilitate read/write operations from the memory of a microprocessor. [4]

(b) Consider the following commands in assembly language for the PICF675 processor:

```
        movlw 0xAA
        movwf 0x35
        decf 0x35
label   decfsz 0x35
        goto label
```

Explain the operation of each of the commands.

Calculate the number of cycles taken for the commands to execute.

Calculate the number of cycles taken if, instead, the following commands are executed:

```
        movlw 0xAA
        movwf 0x35
label   decf 0x35
        decfsz 0x35
        goto label
```

[6]